Harini P

EMBEDDED SYSTEMS | IoT | MICROCONTROLLER DEVELOPMENT

Erode, Tamil Nadu, India | +91 79047 57885 | harinip0705@gmail.com | linkedin.com/in/harini_p0705

CAREER OBJECTIVE

Electronics and Communication Engineering student with a strong interest in **Embedded Systems, IoT, and Microcontroller Programming**. Hands-on experience with **ESP32, Arduino, sensor interfacing, and C programming** through academic projects and an internship. Seeking an entry-level Embedded Systems role to apply and further develop skills in embedded software and hardware design.

TECHNICAL SKILLS

Programming	C Programming, Python (Basic)
Embedded Systems	ESP32, Arduino, GPIO, Sensor Interfacing, Microcontroller Programming
Communication Protocols	UART, I^2C
Operating Systems	Linux (Basic)
Embedded Concepts	Digital Electronics, Basic Embedded Systems, Hardware Interfacing
Tools	Arduino IDE, Wokwi, ThingSpeak

INTERNSHIP EXPERIENCE

QMOS Technologies
Embedded Systems & IoT Intern

- Worked with **Arduino and ESP32**, interfacing sensors and integrating IoT devices with cloud platforms for real-time data monitoring.

PROJECTS

Smart Alert System

- Designed and simulated an **IoT-based environmental safety monitoring system** using ESP32 to detect gas leakage, motion, temperature, and humidity using multiple sensors.
- Integrated cloud monitoring through **ThingSpeak** with real-time alerts using LED and buzzer for safety notifications.

TNEA TFC Zone Mapping Automation System

- Built a prototype desktop application to automate **TNEA TFC data processing** and zone-wise student allocation.
- Supported multiple Excel formats, automatic table detection, and automated report generation.
- **Award:** Secured **1st Prize** in the Computer Science Internal Hackathon at GCEE.

EDUCATION

Government College of Engineering, Erode — 2023 – 2027
B.E. – Electronics and Communication Engineering — CGPA: 7.74

SVN Matric Hr. Sec. School
HSC – Bio-Maths — 2023

CERTIFICATIONS

- **Digital System Design** — NPTEL
- **C Programming** — NPTEL